

Bo Yang

New York, NY

Email: byang0845@gmail.com

Mobile: 860-729-0003

Website: <https://by364.dev>

GitHub: <https://github.com/bya0845>

EXPERIENCE

WSP

NYC

Data Engineer/Structural Engineer

March 2026 – Present

- Build and maintain ETL pipeline for structural engineering group using Azure, Snowflake, Dagster
- Extract, ingest, process data from field inspections, cloud databases, web API. Load data into columnar storage/Data Lake for OLAP queries. Set up BI for business analytics
- Build in-house software to assist with processing bridge inspection measurements and paperwork for NY Thruway bridge inspections

INGENIMAX

REMOTE

Engineering Intern

Dec 2024 – Jan 2025

- Contributed to a platform-engineering AI agent MVP that automated deployment for Amazon EKS clusters based on client needs.
- Tested and evaluated LLM query handling, chain-of-thought reasoning, and context retrieval mechanisms.

SOUTH COL ENGINEERING, P.C.

ALBANY, NY

Structural Engineer

Aug 2021 – Jan 2026

- Conduct bridge inspections in NYSDOT Region 8 (Lower Hudson)
- Take field measurements, gather inspection field data, and perform structural loss calculations. Maintain and update NYSDOT bridge database (BDIS).

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY

DEC 2025

M.S. Computer Science

- GPA: 3.90
- Select Coursework: System Design in Cloud Computing, Graduate Algorithms, Advanced Operating Systems, Deep Learning, Artificial Intelligence, Database System Implementation

UNIVERSITY OF CONNECTICUT

B.S.E. Civil Engineering

PERSONAL PROJECTS

IN-MEMORY MAPREDUCE

Distributed Systems Project

github.com/bya0845/rust-spark

- Built a distributed Spark/MapReduce framework in Rust supporting parallel data transformations and partitioning across worker nodes.
- Supports local deployment via KIND and cloud deployment on Azure AKS.
- Implemented leader election with etcd and inter-node coordination using gRPC and Tokio.

BRIDGE DATABASE MANAGEMENT APPLICATION WITH AI SEARCH

Full-Stack Application

bridge-app.by364.dev

- Built and deployed an AI database search application using Flask and Fly.io.
- Processed individual bridge location and structural data into an SQLite database and implemented natural-language AI search using a lightweight transformer. SQL querying also supported.

SKILLS

Programming Python, Rust, C/C++, SQL

Tools and Frameworks PyTorch, MLflow, pandas, NumPy, Polars, Azure (AKS, AI, Database, Data lake, Spark, Databricks, Web app, Storage), Docker, Kubernetes, CI/CD, Git, gRPC, etcd, MPI, OpenMP, TypeScript

Languages English (native), Mandarin Chinese (native)